

A PROJECT REPORT ON Wine Quality Prediction Using Ridge, Lasso, and Elastic Net Regression

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by

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Project Title:

Wine Quality Prediction Using Ridge, Lasso, and Elastic Net Regression

TEAM NUMBER - 34

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1. ABSTRACT

Predicting wine quality from physicochemical laboratory measurements is a well-known regression problem in the food and beverage industry, where quality is traditionally judged by human tasters through a slow and subjective sensory panel. This project builds a supervised machine learning pipeline that predicts a wine's quality score directly from measurable chemical attributes such as fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free and total sulphur dioxide, density, pH, sulphates, and alcohol content. The core of the project compares three linear regularised regression techniques — Ridge Regression (L2 penalty), Lasso Regression (L1 penalty), and Elastic Net Regression (a convex combination of L1 and L2 penalties) — to understand how each penalty shrinks and, in the case of Lasso and Elastic Net, sparsifies the model coefficients.

The dataset used is the publicly available Wine Quality dataset, which records physicochemical test results alongside a quality score assigned by wine tasters. After cleaning the data and handling missing or inconsistent entries, the numerical features are standardised so that the regularisation penalty is applied fairly across attributes of very different scales. The dataset is split into training, validation, and test partitions, and k-fold cross-validation together with grid search is used to tune the regularisation strength (α) and the Elastic Net mixing ratio ($l1_ratio$) for each model.

Model performance is evaluated using RMSE, MAE, and R^2 , and the three regularised models are further benchmarked against an unregularised linear baseline and a Random Forest regressor to judge whether a simple, interpretable linear model is competitive with a more flexible tree-based ensemble. Principal Component Analysis is additionally applied to visualise correlation structure among the chemical attributes before modelling. The project's key contribution is a practical, side-by-side comparison of how L1 and L2 penalties behave on a real, moderately noisy tabular dataset — showing which chemical attributes Lasso eliminates entirely, which ones Ridge merely shrinks, and how Elastic Net balances the two. The resulting model and its evaluation pipeline demonstrate how regression-based machine learning can support faster, data-driven, and consistent wine quality assessment.

2. INTRODUCTION

The wine industry relies heavily on sensory evaluation panels to assign a quality score to each batch produced. This process is time-consuming, expensive, and subject to the individual biases of human tasters. Machine learning offers an alternative: since wine quality is closely related to its underlying physicochemical composition, a regression model trained on laboratory-measured attributes can learn to predict quality scores directly from data, offering a fast, repeatable, and objective screening tool that can support — not replace — human tasting panels.

This project frames wine quality prediction as a supervised regression task and focuses on regularised linear models: Ridge Regression, Lasso Regression, and Elastic Net Regression. Regularisation is essential here because several physicochemical attributes in the dataset are correlated with one another (for example, fixed acidity and pH, or free and total sulphur dioxide), which can make an ordinary least-squares model unstable and prone to overfitting. Ridge Regression adds an L2 penalty that shrinks all coefficients smoothly and handles correlated features gracefully. Lasso Regression adds an L1 penalty that can shrink some coefficients exactly to zero, effectively performing automatic feature selection. Elastic Net combines both penalties, offering a middle ground that retains groups of correlated features while still encouraging sparsity.

Before modelling, the raw dataset is cleaned and preprocessed: missing values are handled, numerical attributes are standardised using a scaler so that the regularisation penalty is not dominated by attributes with larger numeric ranges, and the data is split into training, validation, and test sets. Hyperparameter tuning is carried out using grid search combined with k-fold cross-validation to select the regularisation strength for each model. The tuned models are then evaluated on a held-out test set using RMSE, MAE, and R^2 , and compared against a plain linear regression baseline and a Random Forest regressor to place their performance in context.

The remainder of this report maps the techniques used in the project to the course modules (COs) from which they are drawn, lists the software requirements, and outlines the methodology followed from raw data to a trained and evaluated model.

3. CO-WISE MODULES COVERED

The project draws a small, focused set of techniques from each module (Course Outcome) of the Machine Learning syllabus, rather than restricting itself to a single module. The mapping below shows exactly which topics were used and where.

CO1 — Module 1: The ML Lifecycle

- The project is treated as a full ML lifecycle exercise: raw dataset → cleaned and standardised data → trained regularised regression model → evaluated and packaged model artifact.
- Wine quality prediction is identified explicitly as a **supervised regression** problem, distinguished from the unsupervised and reinforcement-learning categories introduced in this module.

CO2 — Module 2: Supervised Learning

- Core focus of the project: **Ridge (L2), Lasso (L1), and Elastic Net** regularised regression, including the geometric intuition for why the L1 penalty drives coefficients to exactly zero while the L2 penalty only shrinks them.
- Feature scaling and standardisation are applied before training, since regularised regression is sensitive to the scale of input features.

- Missing-value handling (imputation) is applied during preprocessing of the raw physicochemical measurements.

C03 — Module 3: Tree-Based Models

- A **Random Forest Regressor** is trained as a non-linear benchmark to compare against the regularised linear models.
- Feature importance (permutation importance) from the Random Forest is used to cross-check which chemical attributes Lasso selects versus which ones the tree ensemble finds most predictive.

C04 — Module 4: Unsupervised Learning

- Principal Component Analysis (PCA) is used, after standardisation, to visualise correlation structure among the physicochemical attributes before regression modelling.

C05 — Module 5: Model Evaluation

- A disciplined **train / validation / test split** is used, with the test set held out from every modelling decision.
- k-fold cross-validation is used during hyperparameter tuning of the regularisation strength (alpha) and the Elastic Net mixing ratio (l1_ratio).
- Regression metrics — **RMSE, MAE, and R^2** — are used to evaluate and compare all models.
- Grid search is used for hyperparameter tuning of each regularised model.

C06 — Module 6: ML Engineering

- The final tuned model is packaged in a portable format, with a brief discussion of how it could be served through a lightweight REST API (for example, FastAPI) as a practical next step beyond this course project.

4. METHODOLOGY

- **Data collection:** Use the publicly available Wine Quality dataset (physicochemical tests and quality score) in CSV format.
- **Preprocessing:** Handle missing values, remove duplicate records, and standardise numerical features using a scaler.
- **Splitting:** Divide the data into training, validation, and test sets; apply k-fold cross-validation on the training set.
- **Model training:** Fit Ridge, Lasso, and Elastic Net regression models; tune alpha and l1_ratio using grid search.
- **Baseline comparison:** Compare against an unregularised linear regression baseline and a Random Forest regressor.
- **Evaluation:** Measure RMSE, MAE, and R^2 on the held-out test set; inspect which coefficients Lasso sets to zero.
- **Interpretation:** Rank physicochemical attributes by their influence on predicted wine quality across the three models.

5. SOFTWARE REQUIREMENTS

- **Operating System:** Windows 10/11 or Linux
- **Programming Language:** Python 3.x
- **Development Environment:** Jupyter Notebook or Google Colab
- **Libraries:** NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
- **Dataset:** Wine Quality dataset (CSV format)

These tools are sufficient to preprocess the data, train and tune the Ridge, Lasso, and Elastic Net regression models, benchmark them against baseline models, and visualise and compare their results.

6. EXPECTED OUTCOME

The project is expected to produce a comparative study showing which regularisation strategy — Ridge, Lasso, or Elastic Net — best predicts wine quality from physicochemical attributes, along with a clear picture of which chemical properties (such as alcohol content, volatile acidity, and sulphates) are most influential in determining quality. The resulting pipeline demonstrates a complete, reproducible workflow from raw tabular data to a tuned and evaluated regression model.

Signature of the Guide

Course Instructor